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Temperature response functions for climate impact assessments: The case of ENSO and electricity consumption ☆

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ABSTRACT

We propose a simple regression-based approach to creating counterfactual temperature responses based on high-frequency temperature observations and low-frequency economic responses generated by way of a nonlinear temperature response function (TRF) or cross-TRF interacted with covariates. Our approach allows identification of components of temperature or other climate variables and hence the economic damages attributable to specific climate forcings or sources of natural variability. We apply this methodological approach to estimating the impact of the El Niño Southern Oscillation (ENSO) on electricity consumption in Washington state. By comparing the responses to actual temperatures with those to the counterfactual temperatures, we find that La Niña raises and El Niño reduces electricity consumption, and hence the effect of ENSO alternates between cost and saving, which may be as high as 8.5% per month in a given month. Averaging costs and savings since 1990, ENSO presents a very small net saving (<0.001% per month) in electricity consumption in the state.

1. Introduction

The last two decades have seen a proliferation of studies on the economic impact, usually in the form of damages or more explicitly costs, of climate and climate change. Early influential studies include Schlenker and Roberts (2006, 2009), Deschênes and Greenstone (2007, 2011), Dell et al. (2012), Burke et al. (2015), Hsiang et al. (2017), and see also Howard and Sterner (2017) for a meta-analysis of early studies and Cui et al. (2024, Table 1) for a recent catalog of functions used by studies in leading economics journals to map temperatures to economic damages. Understanding impacts of climate change is a key to calculating a social cost of carbon (Auffhammer, 2018, e.g.) often used for climate policy analysis.

One cost of climate change noted by Hsiang et al. (2017) is an increase in electricity demand, primarily due to cooling needs in the presence of rising temperatures. Indeed, cooling and heating are among the largest uses of electricity, as shown for the US in our Fig. 2 based on data from the Energy Information Administration. Long before rising temperatures were a concern to mainstream economists, energy price increases in the 1970s spurred a literature on energy efficiency measured through a *temperature response function* (TRF). Engle et al. (1986) provided an early influential study on the use of temperature response functions either to estimate or to forecast electricity consumption, and such studies have since proliferated to include many different specifications and many different electricity markets.[1]

Older temperature response function models are typically specified as linear in heating and cooling degree days or piecewise linear in temperature, because we expect to see a V-shaped function due to increasing demand for electricity to heat or cool as temperatures get colder or hotter. The minimum of the TRF reflects a comfort zone or balance point at which neither heating nor cooling is needed. However,

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[1] Examples include Australia (Fan and Hyndman, 2011), Europe (Bessec and Fouquau, 2008), Israel (Beenstock et al., 1999), Mexico (Chang and Martinez-Chombo, 2003), Saudi Arabia (Al-Zayer and Al-Ibrahim, 1996), South Korea (Chang et al., 2014, 2016; Miller and Nam, 2022), Spain (Valor et al., 2001; Pardo et al., 2002; Moral-Carcedo and Vicéns-Otero, 2005), Turkey (Halicioglu, 2007; Arisoy and Ozturk, 2014), the UK (Henley and Peirson, 1998), and the US (Sailor and Muñoz, 1997), but this list is by no means exhaustive. See also the reviews by Brown et al. (2016) and Fazeli et al. (2016). Most notably relevant to our analysis of Washington state, Dubin (2008) estimated the empirical distribution of balance point temperatures based on his engineering thermal model and thermostat settings of individual buildings in the Puget Sound Energy service territory.

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even the early work of Engle et al. (1986) moved beyond piecewise linear specifications embodied by degree days, and the majority of studies since then have use various nonlinear and either nonparametrically or semiparametrically specified and estimated TRFs.

Although the idea that economic covariates, such as price, and especially weather covariates, such as humidity, may affect electricity is well established – consider the familiar heat index, e.g. – Chang et al. (2016) introduce the concept of a cross-TRF to interact these effects with a semiparametrically specified TRF. The cross-TRF approach provides an extremely flexible way to assess how such covariates affect the *shape* and not just the level of the TRF as an additive covariate would. Our TRF modeling approach closely follows the cross-TRF approach of Chang et al. (2016) and Miller and Nam (2022), albeit for a different region and a very different aim.

We employ a TRF not to measure the costs (and perhaps surprising benefits) of temperature change directly, but rather to measure those associated with a particular component of temperature change. Temperatures may be driven by many different factors that operate at different time-scales and frequencies. We may think of the temperature at any one time as being driven by the sum of these various component drivers. The most obvious ones are those with the largest amplitudes, highest frequency, and clearest periodicity: daily and seasonal. Aggregation is one way that researchers may deal with high-amplitude and high-frequency fluctuations such as diurnal cycles. Indeed, when consumption is observable only monthly, aggregation also solves a so-called mixed-frequency problem because temperatures observations may be available at a much higher frequency. However, semiparametric mixed-frequency modeling approach improves statistical efficiency and allows high-frequency temperature variation to inform consumption estimates by tackling this mismatch in frequencies directly.

Other components of temperature may be more subtle and are certainly understudied in the context of electricity consumption modeling, though not necessarily in the context of economic damages from climate change. From the point of view that local or global temperature is driven by a physical mechanism embodied by an energy balance model (Budyko, 1969; Sellers, 1969), climate modelers often decompose temperature change into so-called *climate forcings* and *natural variability*.[2] Climate forcings may be anthropogenic or natural but are defined by anything that moves the climate system away from equilibrium, while natural variability refers to dynamics that are built into the climate system itself. Perhaps the most well-studied forcings are anthropogenic and especially forcing due to an accumulation of carbon dioxide emitted into the atmosphere.

The economic effects of natural variability are less well-known, *except* when the natural variability translates into extreme and damaging weather events such as floods, heat waves, tornadoes, hurricanes, etc. One source of natural variability that does indeed drive such events is the quasi-periodic El Niño Southern Oscillation (ENSO), which consists of hotter El Niño and colder La Niña episodes.[3] ENSO has a period of roughly 5–6 years, with a so-called super El Niño episode every 15 years or so.

Because of the role of ENSO not only in extreme temperatures but also in the movement of moisture that drives extreme weather, damages resulting from ENSO-induced weather is indeed one of the most well-studied climate-driven damages. Studies of damages from ENSO predominantly focus on precipitation and flooding (Magilligan and Goldstein, 2001; Chowdhury, 2003; Davey et al., 2014; Ward et al., 2014a,b, 2016; Munoz and Dee, 2017), crop yields or prices (Tack and Ubilava, 2013; Ubilava and Holt, 2013; Iizumi et al., 2014; Hsiang and Meng, 2015), economic growth in general (Smith and Ubilava, 2017; Ubilava et al., 2019; Callahan and Mankin, 2023; Liu et al., 2023), or even in civil strife (Hsiang et al., 2011).

To the best of the authors' knowledge, little – if any – research has been done specifically linking ENSO to electricity consumption. The link is complicated both by the nonlinearity discussed above but also by the quantification of the component of a temperature increase or decrease that may be attributable to ENSO. We propose a simple regression-based approach to identify such a component. Once we have identified this component, we propose a methodology to generate counterfactual electricity consumption under a hypothetical scenario without ENSO using a cross-TRF specification similar to that of Chang et al. (2016) and Miller and Nam (2022). The counterfactual allows us to estimate costs (damages) associated specifically with ENSO. Although we focus on ENSO, the methodology could be applied in a straightforward way to any other component of temperature or other climate variables such as precipitation.

We focus on the Pacific Northwest (PNW) region of the United States, and Washington (WA) state in particular. Our geographical focus is born out of the well-known effects of ENSO on locations around the Pacific, as well as a somewhat unexpected (relative to the global average) effect of ENSO on the Northern US: while most studies focus on the considerable costs (damages) due to ENSO, we find a small net *saving* effect of ENSO for residential and commercial electricity users in WA once we average costs and savings since 1990.

The savings come from that fact that El Niño effects in the PNW tend to occur during heating rather than cooling months, a pattern that we may expect across more states in the Northern US and possibly Japan, but which we would expect to be reversed in the Southern Hemisphere or in regions where the effect of ENSO on temperature has the opposite sign.[4] While the geographic heterogeneity of ENSO's effects may make our empirical results difficult to generalize directly, the methodology we employ is immediately generalizable to any region of the world for which temperature and electricity consumption data are available.

The rest of the paper is structured as follows. We present the models and methods we use to estimate them in Section 2. We present the data and empirical results in Sections 3 and 4 respectively. Section 5 concludes.

2. Models and econometric methods

The key to estimating cost (or damage) associated with any temperature fluctuations is a damage function such as the temperature response function (TRF) of Engle et al. (1986) and subsequent authors. Specifically, we utilize a cross-TRF along the lines of those of Chang et al. (2016) and Miller and Nam (2022) to allow for covariates. We also need a method for constructing counterfactual costs without ENSO. The difference between the costs with ENSO and the counterfactual costs without ENSO constitute our estimates of economic damages due to ENSO.

[2] A handbook chapter by Castle et al. (2024) on macroeconomic forecasting using climate models provides a treatment of climate models aimed at economists.

[3] Other well-known climate cycles include the Atlantic Multidecadal Oscillation, and Pacific Decadal Oscillation as well as the intra-annual Madden-Julian Oscillation. We acknowledge but do not model interactions between these cycles and ENSO or between anthropogenic forcing and ENSO (Cai et al., 2015, 2018).

[4] The interested reader is referred to global maps of that show the spatially heterogeneous effects of ENSO:

•https://climateknowledgeportal.worldbank.org/country/united-states/enso
•https://www.climate.gov/news-features/featured-images/global-impacts-el-nino-and-la-nina

The first shows a positive correlation between temperatures and the ENSO index that affects Washington, Oregon, North Dakota, Minnesota, and Wisconsin, largely bypassing Montana. Albeit harder to view the states precisely, the second suggests that El Niño affects Washington, Northern Idaho, Montana, and maybe North Dakota. Washington appears to be the common denominator.

2.1. Models

2.1.1. TRF and cross-TRF models

A cross-TRF model for temperature and energy consumption measured at contemporaneous time increments (Miller and Nam, 2022, e.g.) may be written as

$$y_t = \sum_{k=0}^{K} w_{kt} g_k(r_t) + \varepsilon_t, \tag{1}$$

for $t = 1, \ldots, T$ generic time periods, where y_t is a measure of local energy consumption, often detrended in some manner to account for otherwise unmodeled long-run changes in consumption, and r_t is the observed local temperature, which may also be detrended to remove long-run effects from climate change in order to focus on short-run variations in temperature.

By construction, $w_t^0 = 1$ for all t, so that the cross-TRF model reduces to the more familiar TRF model with no covariates when $K = 0$. More generally, w_t^k for $k = 1, \ldots, K$ are K covariates that may affect the response of electricity consumption to temperature. For example, the cross-TRF of Chang et al. (2016) focuses on long-run economic covariates (price and time), while that of Miller and Nam (2022) focuses on short-run weather covariates (humidity, cloud cover, and wind speed).

To simplify subsequent exposition on the estimation of the model, what we refer to as *temperature* will be temperature that has been normalized such that $r_t = (\text{temperature}_t - a)/(b-a)$ with the range of raw temperatures (temperature_t) observed over the entire sample period or with outliers omitted and denoted by $[a, b]$. We may easily redefine g_k as a function of the raw temperature instead of the normalized temperature.

Similarly to Chang et al. (2016) but in contrast to Miller and Nam (2022), we have high-frequency (hourly) temperature data but only low-frequency (monthly) consumption data. We follow the former authors, rewriting the cross-TRF as

$$y_t = \sum_{k=0}^{K} w_{kt} \int_0^1 g_k(r) dF_t(r) + \varepsilon_t, \tag{2}$$

where F_t is the cumulative distribution function of hourly temperatures within month t.

The cross-TRF model in either Eq. (1) or Eq. (2) may be written more succinctly as

$$y_t = w_t' g_t + \varepsilon_t \tag{3}$$

by defining w_t as a $(K+1) \times 1$ vector $(1, w_t^1, \ldots, w_t^K)'$ and $g(z) = (g_0(z), \ldots, g_K(z))'$ for generic argument z, so that either $g_t = g(r_t)$ for the specification in Eq. (1) or else $g_t = \int_0^1 g(r) dF_t(r)$ for the specification in Eq. (2).

Finite second moments are assumed for all series such that $\mathbf{E}|y_t|^2 < \infty$, $\mathbf{E}\|w_t\|^2 < \infty$, and $\mathbf{E}\|g_t\|^2| < \infty$. We assume that $\mathbf{E}[y_t | w_t, g_t] = w_t' g_t$ so that $\mathbf{E}[\varepsilon_t | w_t, g_t] = 0$, or that the model estimates the mean electricity consumption y_t conditional on the current temperature and covariates. We use a residual-based autoregressive bootstrap because we find that the error is serially correlated, but we do not make any allowance for heteroskedasticity in estimating regression confidence intervals.

2.1.2. Counterfactual modeling

We decompose a specific value of r given by r_* into two (or more) orthogonal components r_E and r_0 such that $r_* = r_E + r_0$. Specifically, r_0 denotes a counterfactual temperature without any contribution from ENSO and r_E denotes any component of temperature that is driven by ENSO. Define ∇_E to be an increment driven by ENSO, so that $\nabla_E r_* = r_E = r_* - r_0$. We wish to assess the effect of ENSO on electricity consumption, which we will denote by $w_t' \nabla_E g_t$ assuming that $\nabla_E w_t = 0$, or that the covariates are orthogonal to r_E. We will set aside w_t momentarily and examine $\nabla_E g_t$, the effect of ENSO on g_t.

For a given month t, we observe a distribution of hourly temperatures, and we wish to identify a counterfactual distribution of temperatures. We do so simply by shifting the mean by the amount of the ENSO proxy r_{Et}, which is observed only monthly. Specifically, we define the counterfactual distribution F_{0t} such that $F_{0t}(s) = F_{*t}(r - r_{Et})$ so that

$$r_{*t} = \int_0^1 r\, dF_{*t}(r) = \int_{-r_{Et}}^{1-r_{Et}} (s + r_{Et}) dF_{0t}(s) = \int_{-r_{Et}}^{1-r_{Et}} s\, dF_{0t}(s) + r_{Et} = r_{0t} + r_{Et}$$

by a change of variables and for the monthly average temperature $r_{*t} = \int_0^1 r\, dF_{*t}(r)$ and counterfactual temperature $r_{0t} = \int_{-r_{Et}}^{1-r_{Et}} s\, dF_{0t}(s)$. It follows that

$$\nabla_E r_{*t} = r_{*t} - r_{0t} = \int_0^1 r\, dF_{*t}(r) - \left[\int_0^1 r\, dF_{*t}(r) - r_{Et} \right] = r_{Et}$$

as expected.

For the (cross-)TRF g in Eq. (1) in which all series are observed at the same frequency t,

$$\nabla_E g(r_{*t}) = g(r_{*t}) - g(r_{0t}) = g(r_{*t}) - g(r_{*t} - r_{Et}), \tag{4}$$

where the first equality follows by definition and the second equality allows a calculation based on observable r_{*t} and r_{Et}.

For the (cross-)TRF g in Eq. (2) in which temperatures are observed hourly,

$$g_{0t} = \int_{-r_{Et}}^{1-r_{Et}} g(s) dF_{0t}(s) = \int_0^1 g(r - r_{Et}) dF_{*t}(r)$$

so that

$$\nabla_E g_{*t} = g_{*t} - g_{0t} = \int_0^1 [g(r) - g(r - r_{Et})] dF_{*t}(r) \tag{5}$$

follows by substitution and collecting terms. The expression in Eq. (5) simply means that we calculate the difference between the TRF at each actual hourly temperature and each counterfactual hourly temperature and then weight these differences by the actual distribution of temperatures. Note the difference between $\nabla_E g(r_{*t})$ in Eq. (4) and $\nabla_E g_{*t}$ in Eq. (5). The former evaluates the TRF as a function of monthly aggregates, while the latter evaluates the TRF as an aggregate of functions of high-frequency (hourly) data.

We may construct a counterfactual series of electricity consumption by subtracting w_t' times Eq. (4) or (5) from the series of actual electricity consumption, depending on whether or not we observe high-frequency temperatures. For the low-frequency only case, the counterfactual is constructed as

$$y_{0t} = y_t - w_t' \nabla_E g(r_{*t}) = y_t - w_t'[g(r_{*t}) - g(r_{*t} - r_{Et})], \tag{6}$$

while the counterfactual is constructed as

$$y_{0t} = y_t - w_t' \nabla_E g_{*t} = y_t - w_t' \int_0^1 [g(r) - g(r - r_{Et})] dF_{*t}(r), \tag{7}$$

when we observe high-frequency temperatures. Note that if $\nabla_E w_t' g_{*t} > 0$, the cost is positive (a damage) so the counterfactual is a lower level of electricity consumption. On the other hand if $w_t' \nabla_E g_{*t} < 0$, the cost is negative, so that "damage" is a misnomer and this case could be characterized more aptly as a "saving".

2.2. Econometric methods

Estimating damages from ENSO proceeds in three steps. We first estimate the distribution of hourly temperature anomalies in each month $F_t(z)$. We then estimate the relationship between local electricity consumption and local temperature with additional covariates using the cross-TRF $g(z)$ in Eq. (2) or (3). Finally, we estimate the component r_{Et} of local temperature that is attributable to ENSO. Based on estimates of $F_t(z)$, $g(z)$, and r_{Et}, it is straightforward to estimate the economic damage given by w_t' times Eq. (5) and counterfactual electricity consumption given by Eq. (7).

2.2.1. Step 1: Density estimation

We use a standard kernel density estimator to obtain estimates $\hat{f}_t$ of the probably density function f_t corresponding to the cumulative distribution function F_t. Specifically, we use a truncated Gaussian kernel with Silverman bandwidth on the unit interval evaluated at increments of 0.001. We choose the support of the density to capture 99% of the observations, treating the remaining 1% of observation as outliers, along the lines of Chang et al. (2020) and Miller and Nam (2020). Specifically, the support over which we estimate the density is $[a, b] = [-4.775, 29.715]$ °C, which is then normalized to the unit interval. Our estimates are robust to using a greater support that does not exclude outliers and to alternative kernel specifications (results not shown). As do previous researchers, we rely on the large number of hourly observations in each month, 672 (February in a non-leap year) to 744 (months with 31 days), and do not explicitly account for measurement error from this step.

2.2.2. Step 2: TRF and cross-TRF estimation

We employ the Fourier flexible form to approximate and estimate the cross-TRF. The use of this semiparamteric specification in the econometrics literature traces back to Gallant (1981) and is commonly utilized in electricity modeling (c.f., Chang et al., 2014, 2016, e.g.). Consider first a TRF model, or equivalently a cross-TRF model with no covariates – i.e., $w_t^0 = 1$. The approximation to the TRF g consists of polynomial and trigonometric functions given by

$$g_k(z) \approx \sum_{i=0}^{p_k} \beta_{k0i} z^i + \sum_{j=1}^{q_k} [\beta_{k1j} \cos(2\pi jz) + \beta_{k2j} \sin(2\pi jz)] \tag{8}$$

which becomes more precise as p_k and q_k increase.

Miller and Nam (2022) employ hourly data for both temperature and electricity consumption, with the latter from automatic meter reading employed in South Korea, so that they may estimate Eq. (1) with the approximation in Eq. (8) for each hour. In contrast, Chang et al. (2014, 2016) employ monthly consumption data for South Korea and use the inherently mixed-frequency time series model given by Eq. (2) with the approximation in Eq. (8) also for each hour.

Given our mixed-frequency data (monthly electricity consumption and hourly temperature data), our approach to estimation in this step is similar to that of Chang et al. (2014, 2016). Specifically, by substituting the approximation in Eq. (8) into the cross-TRF model in Eq. (2), we may write

$$\begin{aligned} y_t &= \sum_{k=0}^{K} w_{kt} \int_0^1 \left(\sum_{i=0}^{p_k} \beta_{k0i} r^i + \sum_{j=1}^{q_k} (\beta_{k1j} \cos(2\pi jr) + \beta_{k2j} \sin(2\pi jr)) \right) dF_t(r) + \eta_t \\ &= \sum_{k=0}^{K} w_{kt} \begin{pmatrix} \sum_{i=0}^{p_k} \beta_{k0i} \int_0^1 r^i \; dF_t(r) \\ + \sum_{j=1}^{q_k} \beta_{k1j} \int_0^1 \cos(2\pi jr) \; dF_t(r) \\ + \sum_{j=1}^{q_k} \beta_{k2j} \int_0^1 \sin(2\pi jr) \; dF_t(r) \end{pmatrix} + \eta_t, \end{aligned} \tag{9}$$

where we use η_t as a generic error term that captures both the original idiosyncratic error ε_t and the approximation error inherent from finite (p_k, q_k). The approximation error naturally decreases as (p_k, q_k) increase, but we follow the extant literature in choosing (p_k, q_k) to be relatively small to avoid over-fitting the model. We specifically use the Schwarz-Bayesian information criterion (BIC) to choose (p_k, q_k).

Employing estimates $\hat{f}_t$ from the first step, we calculate polynomial and trigonometric functions of hourly temperatures averaged over each month to obtain

$$x_{0it} = \int_0^1 r^i \; \hat{f}_t(r) dr$$

$$x_{1jt} = \int_0^1 \cos(2\pi jr) \; \hat{f}_t(r) dr$$

$$x_{2jt} = \int_0^1 \sin(2\pi jr) \; \hat{f}_t(r) dr$$

for $i = 0, \ldots, p$ and $j = 1, \ldots, q$.

Substitution of x_{0it}, x_{1it}, and x_{2it} into Eq. (9) allows

$$y_t = \sum_{k=0}^{K} w_{kt} \left(\sum_{i=0}^{p_k} \beta_{k0i} x_{0it} + \sum_{j=1}^{q_k} (\beta_{k1j} x_{1jt} + \beta_{k2j} x_{2jt}) \right) + \eta_t, \tag{10}$$

up to approximation error from using $\hat{f}_t(r)dr$ in place of $dF_t(r)$. Note that Eq. (10) is linear in parameters. If we redefine p and q to be the sums of p_k and q_k respectively over $k = 0, \ldots, K$ and define β_{pq} to be the $(1 + p + 2q) \times 1$ coefficient vector and x_t^{pq} to be the $(1 + p + 2q) \times 1$ vector defined by all $p + 1$ x_{0it}'s and all q (x_{1jt}, x_{2jt})'s, we may rewrite the model in Eq. (10) more succinctly as

$$y_t = x_t^{pq\prime} \beta_{pq} + \eta_t, \tag{11}$$

which may be estimated using using least squares.

Similarly to Chang et al. (2016), we expect electricity prices and changing preferences and technology over time to induce changes in the TRF, so we employ cross-TRFs that add interactions between the TRF and these covariates. Specifically, $w^{(k)}$ for $k = 0, \ldots, K = 2$ contains the covariates of concern: a constant to obtain a *base* TRF, time (months), and the retail price of electricity. For convenient interpretation, time is normalized by subtracting the initial year from each year of the sample and dividing by the total number of months in the sample less one, $T - 1 = 407$, so that

$$\{w_t^{(1)}\} = \{(t-1)/(T-1)\} = \{0 = 0/(T-1), 1/(T-1), \ldots, (T-2)/(T-1), (T-1)/(T-1) = 1\}.$$

Prices are standardized by subtracting the sample mean and then dividing by the sample standard deviation. This specification allows the temperature response to vary over time and price.

2.2.3. Step 3: Temperature decomposition and counterfactual analysis

We require an orthogonal decomposition of local temperature given by $r_* = r_0 + r_E$ in order to implement the counterfactual analysis discussed above. Define hourly local temperature $r_{h(t)}$. We may write $r_{h(t)} = (\mu + M_t'\gamma_M + u_{h(t)}) + E_t'\gamma_E$ for a parameter vector $\gamma = (\mu, \gamma_M', \gamma_E')'$ where M_t is an 11×1 vector with 11 monthly dummies and E_t is a vector of one or more proxies for ENSO. The mismatch of frequencies between $r_{h(t)}$ and M_t, E_t simply means that we can only account for monthly rather than hourly fluctuations in local temperature due to ENSO. Hourly fluctuations in $r_{h(t)}$ enter through $u_{h(t)}$.

Of course, the value of γ that contemporaneously orthogonalizes the decomposition is the least squares estimator from a regression of $r_{h(t)}$ onto $(1, M_t', E_t')'$ given by

$$\hat{\gamma} = \left(\sum_{t=1}^{T} (1, M_t', E_t')'(1, M_t', E_t') \right)^{-1} \sum_{t=1}^{T} \left((1, M_t', E_t')' \frac{1}{H_t} \sum_{h=1}^{H_t} r_{h(t)} \right)$$

where H_t is the number of hours in month t. In this light, we decompose hourly local temperature $r_{h(t)}$ as

$$r_{h(t)} = \hat{\mu} + M_t'\hat{\gamma}_M + E_t'\hat{\gamma}_E + \hat{u}_{h(t)} \tag{12}$$

where $\hat{\gamma} = (\hat{\mu}, \hat{\gamma}_M', \hat{\gamma}_E')'$ and $(\hat{u}_{h(t)})$ is an hourly series of fitted residuals from the regression.

The counterfactual y_{0t} and damages $w_t'\nabla_E g_{*t}$ are constructed as in Eq. (7) with monthly $r_{Et} = E_t'\hat{\gamma}_E$ and the monthly distribution of temperatures $F_{*t}(r)$ estimated from the hourly temperatures for each t in the first step. Note that we use actual prices to construct the counterfactual, rather than counterfactual prices or counterfactual consumption due to such prices. We acknowledge the likelihood that counterfactual temperatures would affect prices and hence consumption, but our empirical analysis suggests that prices have only small and in some cases insignificant effects on consumption, which is primarily driven by temperature. If we were to remove the price TRF to average out the temperature response over all prices, we expect that the counterfactual analysis would pick up the transmission of the temperature signal through price indirectly and that our results would be very similar.

Table 1
WA counties and weather stations.

County	Population	City	Station ID	Earliest month
King	2,269,675	Seattle	USW00024233	1990.01
Pierce	921,130	Tacoma	USW00024207	1990.01
			USW00094274	1990.01
Snohomish	827,957	Everett	USW00024222	1990.01
Spokane	539,339	Spokane	USW00024157	1990.01
Clark	503,311	Vancouver	USW00024229	1990.01
Thurston	294,793	Olympia	USW00024227	1990.01
Kitsap	275,611	Bremerton	USW00094263	2005.01
Yakima	256,728	Yakima	USW00024243	1990.01
Whatcom	226,847	Bellingham	USW00024217	1990.01
Benton	206,873	Richland	USI0000KRLD	2015.10
Skagit	129,523	Burlington	USW00094282	2004.05
Cowlitz	110,730	Longview	USW00024223	1990.02
Franklin	96,749	Pasco	USW00024163	1990.01
Grant	99,123	Moses Lake	USW00024110	1990.01
Island	86,857	Oak Harbor	USW00024255	1990.01
Lewis	82,149	Centralia	USW00000119	1990.01
		Toledo	USW00024241	1990.01
Clallam	77,155	Port Angeles	USW00024228	2015.03
Chelan	79,074	Wenatchee	USW00094239	1990.01
Grays Harbor	75,636	Hoquiam	USW00094225	1990.01
Mason	65,726	Shelton	USW00094227	1990.01
Walla Walla	62,584	Walla Walla	USW00024160	1990.01
Whitman	47,973	Pullman	USW00094129	1990.01
Kittitas	44,337	Ellensburg	USW00024220	1990.01
Douglas	42,938	East Wenatchee	USW00094239	1990.01
Okanogan	42,104	Omak	USW00094197	1990.01

25 counties that account for 96.9% of the 2020 population of Washington state, and 26 weather stations in these counties from which we sample temperature data. One station with a sufficiently long record, Pangborn Memorial Airport, lies near the border of Chelan and Douglas Counties, so we use it for both counties. Pierce and Lewis Counties each have two airports with sufficiently long records, so we use two stations for each of those. Pierce County has a substantial airport (Tacoma Narrows) and an airforce base (McChord Air Force Base), so we use both.

3. Data

We obtain monthly electricity consumption for Washington State (WA) from the State Electricity Profiles of the U.S. Energy Information Administration (EIA) from January 1990 through December 2023. The monthly form EIA-861M (formerly EIA-826),[5] includes retail sales (consumption), revenues, customer counts, and retail price by state and sector from 1990 to the present. We focus on the commercial and residential sectors, which together accounted for 76.5% of total consumption in 2022, because they are more susceptible to temperature variability than the industrial sector (Chang et al., 2016) and retail sales to the remaining sector, transportation, are less than 1% of the total. We convert aggregate consumption to per capita consumption by dividing by postcensal estimates of the state population made by the Washington State Office of Financial Management.[6,7]

Hourly local temperatures from 26 land surface stations shown in Table 1 are obtained from the Global Historical Climatology Network (GHCN), a database maintained by the National Centers for Environmental Information (NCEI).[8] The 26 stations reflect those with sufficiently complete records near or in a relatively populous city within the 25 most populous counties, which together account for 96.9% of the 2020 state population. The stations (cities) range over three climate regions, Cfb (Temperate oceanic), Csb (Warm-summer Mediterranean) and Bsk (Cold semi-arid) according to the Köppen climate classification.[9]

Washington's population is concentrated in the Seattle metropolitan area or the greater Puget Sound region. Other populated urban areas include Spokane, Vancouver, and Yakima (see 2023 World Population Review[10]). We match station-level temperatures to statewide electricity consumption by weighting local temperatures in the stations in Table 1 by county population relative to the aggregate population in all 25 counties shown, also from the Washington State Office of Financial Management.[11] The earliest available month for each station is shown in the rightmost column of the table, and we adjust the weights by county data availability so that the county weights sum to one in each year.

The top panel of Fig. 1 shows hourly temperatures from January 1, 1990 through December 31, 2023. Hourly temperatures mostly fall within −10 °C (14 °F) to 35 °C (95 °F). Naturally, the series exhibit strong seasonality, hitting troughs during boreal winters (January-March) and attaining crests during boreal summers (July-September).

For illustration only and at the cost of reduced fidelity, Fig. 1 also shows the monthly average temperature in comparison with monthly electricity consumption per capita for the residential sector (middle panel) and commercial sector (bottom panel). We observe a countercyclical pattern for both sectors. In contrast to many (hotter) locations in the US, peak consumption occurs when the weather is coldest rather than hottest, presumably because winter demand for heating is stronger than summer demand for cooling in Washington. A much smaller local maximum appears to recur every summer since about 2006 in residential consumption, suggesting the growing use of cooling appliances.

Commercial consumption has much smaller yearly variability than residential (note the right y-axis in the bottom panel is downscaled relative to that in the top panel) but shows an upward and then downward trend over time. It has two peaks each year since approximately 2000, corresponding to the peak and trough in temperature. The summer peak has a lower amplitude than winter yet grows with time, gradually catching up with the winter one.

Data from the Annual Energy Outlook 2023[12] at the national level shed some light on why temperature matters. Fig. 2 shows that in 2022 heating, cooling, and water heating ranked as the top three residential end uses. The four largest end uses by the commercial sector were computing and office equipment, refrigeration, cooling, and lighting. Although Washington might diverge in energy uses from the national proportion due to its distinct climate, the narrower variability in commercial consumption is likely because some primary commercial uses, such as office equipment, are less sensitive to seasonal changes in temperature. Cooling, ventilation, and lighting are three uses that plausibly vary with seasons, helping to explain the double peaks in the commercial sector.

A commonly used measure of ENSO in *Niño 3.4*, and we use that based on sea surface temperatures from HadISST1 from Rayner et al. (2003). The series represents the Equatorial sea surface temperature in the region of 5°N–5°S and 170°W–120°W, roughly from the International Dateline through the South American coast. The anomaly is calculated by removing a base period mean (over 1981–2010) from the original series, and the periods dubbed *El Niño* and *La Niña* are defined when the anomaly exceeds +/-0.4 °C for at least six consecutive months. Specifically, El Niño episodes are hotter, while La Niña episodes are colder.

[5] Downloaded from https://www.eia.gov/electricity on December 20, 2024.

[6] Downloaded from https://ofm.wa.gov/washington-data-research on December 20, 2024.

[7] Dividing total consumption by population is less desirable in the commercial sector than in the residential sector. Electricity consumption is determined by the size of the commercial workforce, which is not at a constant proportion of the population over time.

[8] Downloaded from https://www.ncei.noaa.gov/data on December 20, 2024.

[9] https://koeppen-geiger.vu-wien.ac.at/usa.htm

[10] https://worldpopulationreview.com/states/cities/washington

[11] Downloaded from https://ofm.wa.gov/washington-data-research on December 20, 2024.

[12] https://www.eia.gov/outlooks/aeo

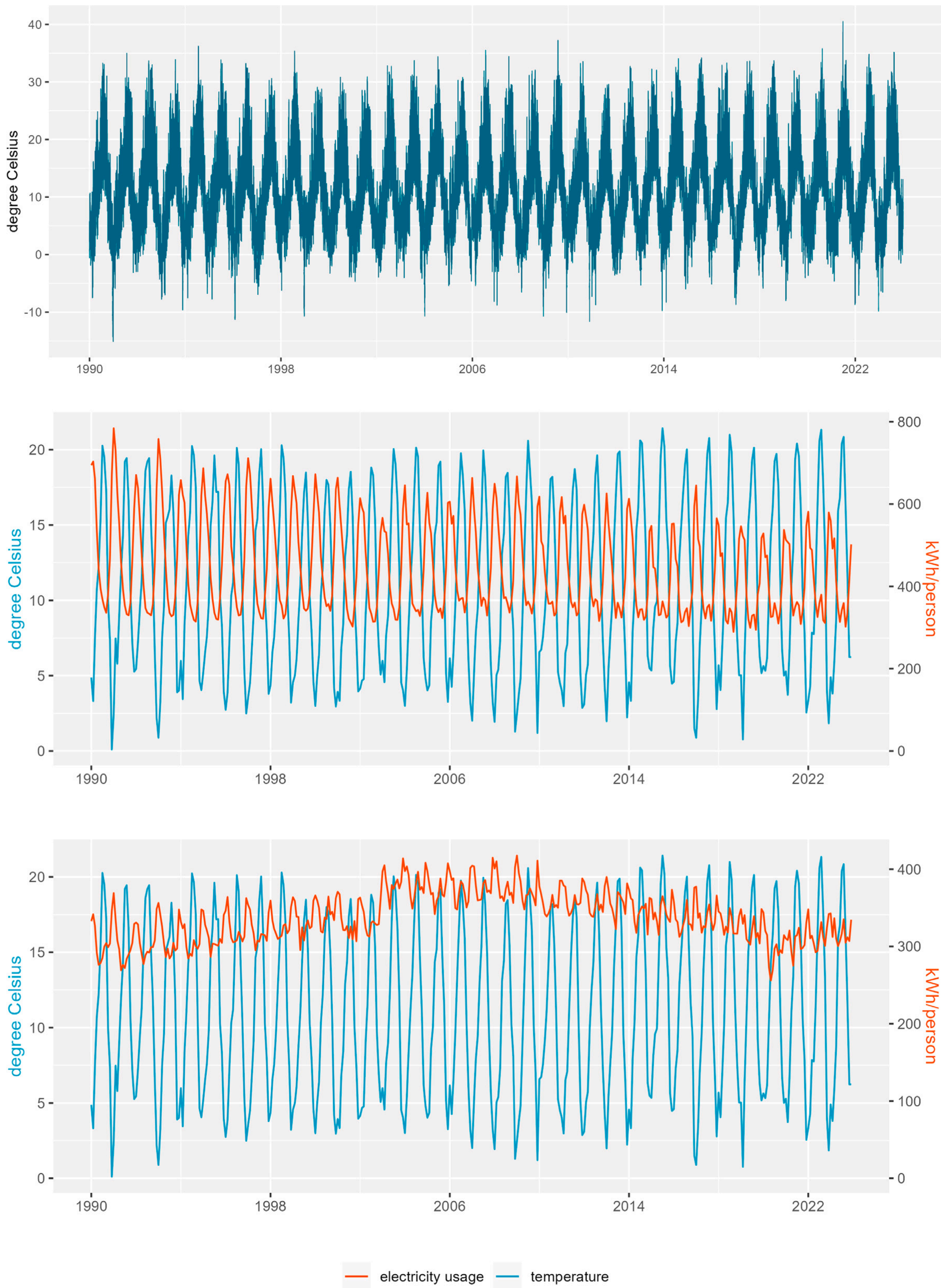

Fig. 1. WA Temperature and Electricity Consumption Per Capita. Spatially aggregated hourly temperature for WA from January 1, 1990 through December 31, 2023 (top panel). Monthly electricity consumption for the residential sector (middle panel) and the commercial sector (bottom panel) compared to the monthly average of hourly temperatures.

Fig. 3 illustrates the Niño 3.4 series that we employ and its temporal characteristics. Looking at the top panel of the figure, the amplitude of the cycle is quite large, as much as ±2.5 °C where it is measured. Its effects are geographically heterogeneous, so the temperature variation

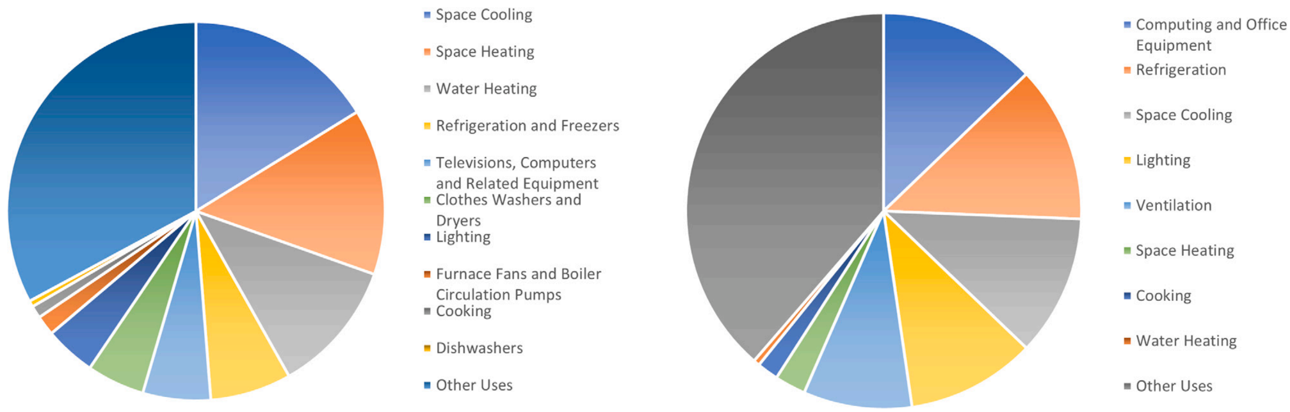

Fig. 2. US Electricity End Uses. End uses for the US in 2022, from EIA Annual Energy Outlook 2023, Tables 4 and 5, for the residential sector (left panel) and commercial sector (right panel). The largest slice of pie in each panel is "Other Uses".

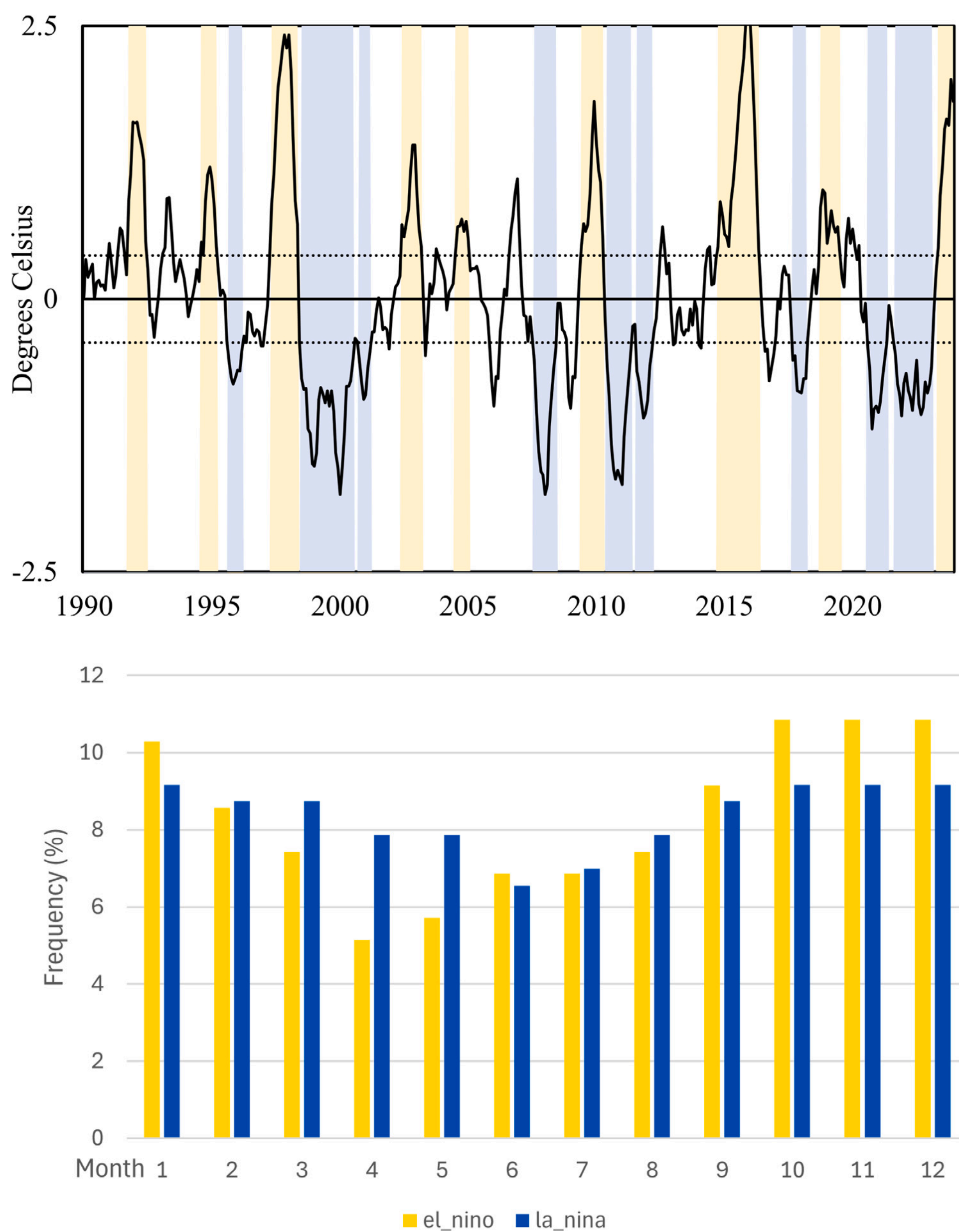

Fig. 3. Niño 3.4 Sea Surface Temperature Anomalies. Top panel: Time series of anomalies. Dotted lines mark ±0.4 °C. Yellow and blue backgrounds represent El Niño and La Niña episodes, respectively. Bottom panel: Distribution of El Niño and La Niña episodes across months of the year.

Table 2
TRF model parameter estimates.

Coefficient	Residential		Commercial	
	Est.	95% Conf. Int.	Est.	95% Conf. Int.
TRF				
β_{000}	6.208	(6.180, 6.235)	5.751	(5.737, 5.766)
β_{011}	0.282	(0.253, 0.313)	0.132	(0.118, 0.147)
β_{021}	0.305	(0.290, 0.319)	0.023	(0.016, 0.031)
Time				
β_{100}	−0.167	(−0.211,−0.122)	0.335	(0.278, 0.390)
β^{*}_{100}			0.423	(0.400, 0.445)
β^{**}_{100}			−0.726	(−0.789,−0.663)
Price				
β_{200}	−0.007	(−0.020, 0.006)	−0.004	(−0.010, 0.002)

Unit: log consumption (kWh per capita). Both models are estimated using least squares on Eqs. (13) and (14) for the residential and commercial sectors respectively. 95% confidence intervals constructed using an autoregressive residual-based bootstrap with 9999 replications.

due to ENSO may be quite a bit different in WA than over the Pacific. Because our goal is to construct a series of counterfactual electricity consumption in the absence of ENSO, it is notable that some years during the sample, specifically 1990, 2006, and 2013 should have less noticeable effect from ENSO. Specifically, those are years in which ENSO was in neither an El Niño nor a La Niña episode, because the +/-0.4 °C threshold was not exceeded.

The bottom panel of Fig. 3 shows the distribution of effects over the course of a year. Hotter El Niño episodes tend to occur more often in boreal falls and winters, specifically October through January, while colder La Niña episodes tend to occur more often in boreal springs, specifically March, April, and May. This seasonal pattern gives us some clues as to the effects of ENSO: El Niño may save electricity in the heating season, while La Niña may boost electricity consumption in volatile spring months. There is effectively no difference in the frequency of either type of episode in the summer months, so ENSO may have very little net effect on cooling demand in the PNW.

4. Empirical results

4.1. TRF estimation

We first estimate a TRF model with additive covariates as a benchmark model in contrast to the proposed cross-TRF model that allows covariates to change the shape of the response of electricity. BIC selects $(p_0, q_0) = (0, 1)$ for both residential and commercial sectors from a maximum order of 3 for each of p_0 and q_0. Table 2 shows the sectoral estimates of the TRF in Eq. (2) with covariates, meaning that K increases beyond zero to allow for covariates, but the lag orders p, q are restricted to zero for $k > 0$ so that the approximation in Eq. (8) reduces to a constant for each $k > 0$.

Specifically, the residential sector TRF is estimated from the regression

$$y_t = \beta_{000} + (\beta_{011}x_{11t} + \beta_{021}x_{21t}) + \beta_{100}((t-1)/(T-1)) + \beta_{200}p_t + \eta_t \quad (13)$$

with standardized retail residential price p_t. The commercial sector TRF is similarly estimated from the regression

$$\begin{aligned} y_t = {} & \beta_{000} + (\beta_{011}x_{11t} + \beta_{021}x_{21t}) + \beta_{100}((t-1)/(T-1)) + \beta_{200}p_t \\ & + \beta^{*}_{100}\mathbf{1}_t(t \geq 2002.11) + \beta^{**}_{100}\mathbf{1}_t(t \geq 2002.11)((t-1)/(T-1)) + \eta_t \quad (14) \end{aligned}$$

with standardized retail commercial price p_t, and where $\mathbf{1}_t(t \geq 2002.11)$ denotes an indicator function for a level shift in November 2002, and where again we use η_t generically for the error term, even though it formally differs between specifications.

Revisiting Fig. 1 provides some impetus for including a trend in the residential sector and a broken trend in the commercial sector.[13] Although there is no obvious long-run pattern in temperatures, there does appear to be a long-run downward trend in residential energy consumption, especially in the peak season. We speculate that this decline reflects an increase in energy efficiency (per capita) over the time span of the data.

The long-run pattern in consumption in the commercial sector is more nuanced. Consumption seems to have trended up until 2002, at which point there is both a level shift and a trend shift in the opposite direction. It is possible that the trend has flattened since the Covid-19 pandemic in 2020, but this is less pronounced. Our research has not uncovered a reason for the evident break in 2002, but we note a break at the same time in the retail price series (not shown). Many states, notably California, deregulated their electricity markets near the turn of the century, but Washington was not one of them. It is well known that the West Coast of the US faced an energy crisis in the fall and winter of 2000 and the spring of 2001. The crisis was the result of the confluence of a number of adverse trends and events.[14] Both small-scale and larger conventional generation were brought online as a result of the crisis, and load was reduced. However, no clear connection is evident between these responses to the crisis and the persistent change in the trend of both prices and consumption in the commercial sector.

Table 2 shows the coefficient estimates from the regressions in Eqs. (13) (residential) and (14) (commercial). The first set of coefficient estimates ($\beta_{0\bullet\bullet}$) govern the TRF, but we also include simple time and price covariates to gauge their importance to each sector. Both the coefficient estimates and confidence intervals for price suggest only a small and statistically insignificant effect in both sectors, similar to the result of Chang et al. (2016) for residential consumption in Korea.

To dig deeper into the residential TRF, Fig. 4 illustrates the residential consumption data plotted against temperature (top panel) and the estimated TRF $\hat{g}(r)$ for the residential sector (bottom panel). By coloring the months in each year in the top panel, we observe an interesting pattern in the residential consumption. Aside from the obvious level effect of seasonality, there is higher demand in earlier months of the year than in later months at similar average temperatures. For example, April and October tend to have similar average temperatures, yet electricity consumption per capita is generally higher in April than in October. The same pattern appears for March and November and, perhaps to a lesser extent, for September and May.

The evidently higher electricity consumption in the spring may very well result from more temperature variation within each spring month than in similar fall months. Moreover, ENSO and particularly La Niña may play a role in this seasonal pattern. Recall from Fig. 3 that March, April, and May have more frequent La Niña episodes. Any increase in intra-monthly temperature volatility due to La Niña or any other cause would indeed drive consumption to be higher even for the same average temperature.

The continuous plot of the residential TRF in the bottom panel of Fig. 4 is the dot product of estimates $(\hat{\beta}_{000}, \hat{\beta}_{011}, \hat{\beta}_{012})$ and

[13] In contrast, Moral-Carcedo and Vicéns-Otero (2005) Bessec and Fouquau (2008), and Chang et al. (2016) work with detrended electricity consumption, but there should not be any major difference between detrending and including the same trend directly in the model. We get similar residuals for each sector if we instead detrend relative to a 12-month moving average as do Chang et al. (2014, 2016) (results not shown). However, we must take care to interpret the time TRFs accordingly.

[14] "[The western electricity crisis] had its roots in several years of under-investment in generating and conservation resources. It was triggered by the onset of poor hydro conditions in the later spring of 2000 leading to the second-worst water year since 1929. It was made much worse by a deeply flawed electricity market design in California and opportunism by some of the participants in that market." Northwest Power and Conservation Council (2005, pg. 3).

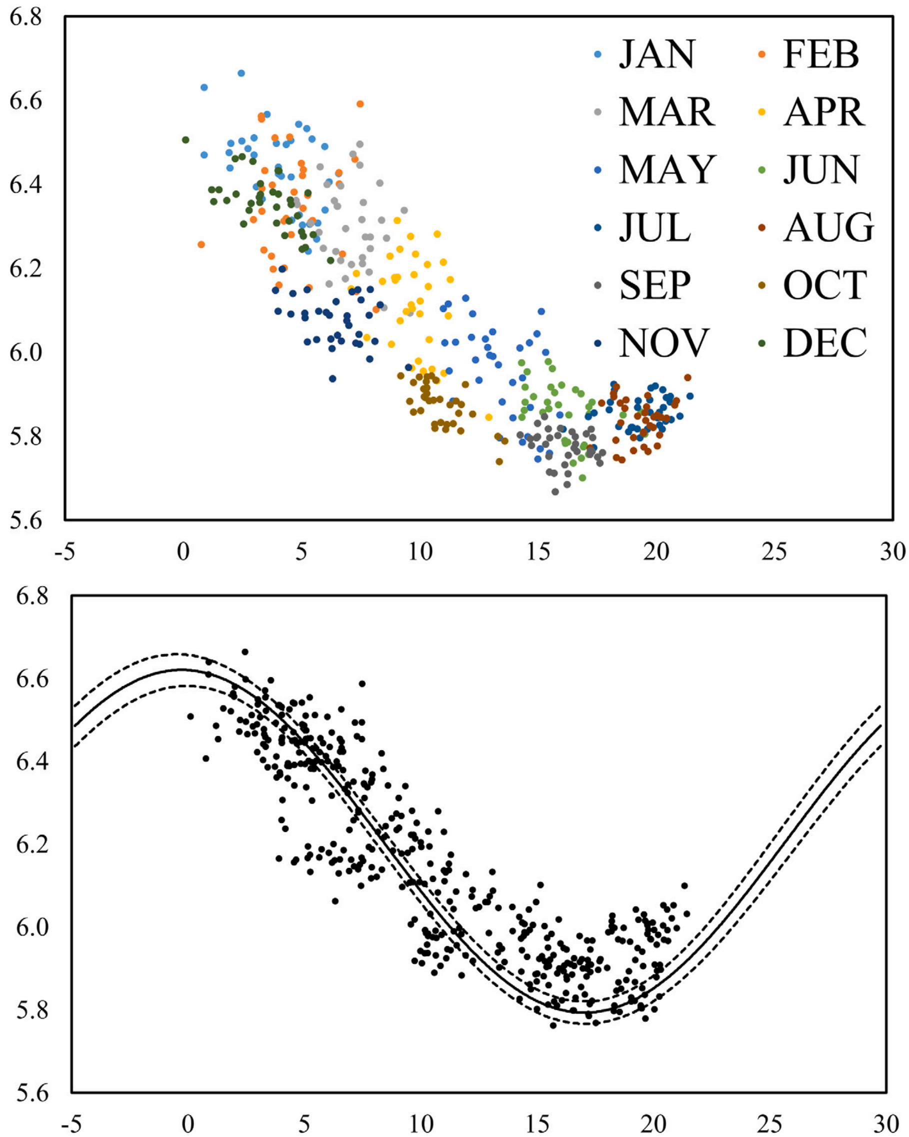

Fig. 4. Residential Consumption: Scatter Plots and Estimated TRF. Unit: log consumption (kWh per capita). Top panel: Monthly residential consumption per average monthly temperature with dot colors denoting different months of each year. Bottom Panel: Monthly residential consumption net of the (linear) effect of time and price per average monthly temperature and residential sector model estimated from Eq. (13) with parameter estimates given in Table 2. 95% confidence bands constructed using an autoregressive residual-based bootstrap with 9999 replications.

$(1, \sin(2\pi r), \cos(2\pi r))$ for r normalized from temperatures on the interval $[-4.775, 29.715]$ °C ($[23.405, 85.487]$ °F) on the horizontal axis. Note that the data plotted here are monthly averages with retail residential price and a linear time trend conditioned out, so while the vertical height of the dots is comparable to the TRF, hourly temperature fluctuations used to estimate the TRF are not shown.

The curve looks like a ladle, whose main part ("handle") goes downward hitting the stationary point on the right ("scoop"). Up to about 17 °C (62.6 °F), people use less electricity in their dwellings as the temperature rises. For instance, a 10-degree increase from 5 °C to 15 °C decreases per capita electricity consumption by about 46%. Electricity consumption picks up again at temperatures exceeding 17 °C. For instance a 10-degree increase from 19 °C to 29 °C increases per capita electricity consumption by about 88%.

Shown in Fig. 5, the commercial sector TRF has a more symmetric U shape, suggesting that cooling plays a bigger role relative to space heating than in the residential sector, as is indeed the case for the US as a whole (Fig. 2). We note, however, that the commercial TRF is flatter than the residential TRF in the sense that the vertical increments for the commercial figure are half those for the residential figure. These U shapes are broadly consistent with the vast literature on temperature response functions for countries of temperate climate zones, such as Germany (Bessec and Fouquau, 2008).[15]

4.2. Cross-TRF estimation

As in the case of the TRF, we use BIC to select the lag orders. Limiting the maximum lag order to 3 entails selecting from $4^6 = 4096$ models. The residential cross-TRF is estimated from the regression

$$y_t = \beta_{000} + \beta_{011}x_{11t} + \beta_{021}x_{21t} + ((t-1)/(T-1))(\beta_{100} + \beta_{111}x_{11t} + \beta_{121}x_{21t}) + p_t(\beta_{200} + \beta_{201}x_{01t}) + \eta_t \tag{15}$$

[15] Seattle (47.6°N) has almost the same latitude as Zürich (47.4°N), though its climate type is more similar to Porto, Portugal (Cfb: warm-summer Mediterranean).

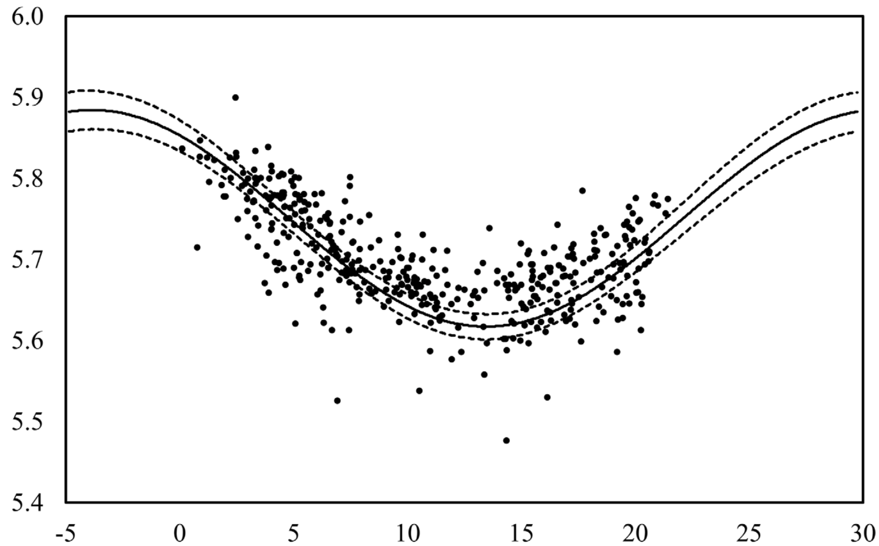

Fig. 5. Commercial Consumption: Scatter Plot and Estimated TRF. Unit: log consumption (kWh per capita). Monthly commercial consumption net of the effect of price and the (broken linear) effect of time per average monthly temperature and commercial sector model estimated from Eq. (14) with parameter estimates given in Table 2. 95% confidence bands constructed using an autoregressive residual-based bootstrap with 9999 replications.

Table 3
Cross-TRF model parameter estimates.

Coefficient	Residential		Commercial	
	Est.	95% Conf. Int.	Est.	95% Conf. Int.
Base TRF				
β_{000}	6.150	(6.113, 6.185)	5.745	(5.725, 5.763)
β_{011}	0.203	(0.142, 0.263)	0.138	(0.110, 0.167)
β_{021}	0.386	(0.346, 0.427)	0.060	(0.043, 0.077)
Time Cross-TRF				
β_{100}	−0.060	(−0.122, 0.003)	0.212	(0.081, 0.346)
β^{*}_{100}			0.424	(0.403, 0.446)
β^{**}_{100}			−0.713	(−0.775,−0.651)
β_{101}			0.254	(−0.319, 0.822)
β_{102}			−0.009	(−0.597, 0.582)
β_{111}	0.146	(0.045, 0.247)		
β_{121}	−0.153	(−0.230,−0.080)		
Price Cross-TRF				
β_{200}	−0.042	(−0.080,−0.002)	−0.014	(−0.029, 0.002)
β_{201}	0.054	(−0.026, 0.134)	0.009	(−0.023, 0.042)

Unit: log consumption (kWh per capita). The residential model has a base TRF with of order $(p_0, q_0) = (0, 1)$ (coefficients $\beta_{0\bullet\bullet}$), a time cross-TRF of order $(p_1, q_1) = (0, 1)$ (coefficients $\beta_{1\bullet\bullet}$), and a price covariate $(p_2, q_2) = (1, 0)$ (coefficients $\beta_{2\bullet\bullet}$). The commercial model has a base TRF with of order $(p_0, q_0) = (0, 1)$ (coefficients $\beta_{0\bullet\bullet}$), a time cross-TRF of order $(p_1, q_1) = (2, 0)$ (coefficients $\beta_{1\bullet\bullet}$), and a price covariate $(p_2, q_2) = (1, 0)$ (coefficients $\beta_{2\bullet\bullet}$). 95% confidence intervals constructed using an autoregressive residual-based bootstrap with 9999 replications.

from BIC choosing $(p_0, q_0) = (0, 1)$, $(p_1, q_1) = (0, 1)$, $(p_2, q_2) = (1, 0)$. On the other hand, the commercial cross-TRF is estimated from the regression

$$
\begin{aligned}
y_t &= \beta_{000} + \beta_{011}x_{11t} + \beta_{021}x_{21t} + ((t-1)/(T-1))(\beta_{100} + \beta_{101}x_{01t} + \beta_{102}x_{02t}) \\
&+ p_t(\beta_{200} + \beta_{201}x_{01t}) + \beta^{*}_{100}\mathbf{1}_t(t \geq 2002.11) \\
&+ \beta^{**}_{100}\mathbf{1}_t(t \geq 2002.11)((t-1)/(T-1)) + \eta_t \qquad (16)
\end{aligned}
$$

from BIC choosing $(p_0, q_0) = (0, 1)$, $(p_1, q_1) = (2, 0)$, $(p_2, q_2) = (1, 0)$. Note that we repeat the broken trend from the commercial sector TRF in Eq. (14). Aside from the broken trend, the difference between these specifications lies in the time TRF in Eqs. (15) and (16). The former (residential) includes a pair of trigonometric functions, like the base TRFs, while the latter (commercial) includes a quadratic. We present the coefficient estimates of the cross-TRF models in Table 3.

The base TRFs and cross-TRFs are displayed in Figs. 6 and 7 for the residential and commercial sectors respectively. In the simple TRF model of the previous section, we could interpret the TRF as $\mathbb{E}(y_t|F_t)$ or energy consumption given the temperature distribution. The interpretation of the base TRF with cross-TRFs is a bit more nuanced, because the base TRF $\int g_0(r)dF_t(r)dr$ is $\mathbb{E}(y_t|F_t; w_1, w_2 = 0)$ or energy consumption given the temperature distribution *when the covariates are zero*. Normalizing time to the unit interval and standardizing price, as we have done, facilitates interpretations of the TRFs. Specifically, $((t-1)/(T-1)) = 0$ corresponds to the beginning of the sample, January 1990, and $p_t = 0$ corresponds to the sample average retail price in the respective sector.

In fact, the base TRF shown in Fig. 6 for the residential sector is comparable to the TRF for the residential sector in Fig. 4, except that the TRF in Fig. 4 reflects an average over the whole sample, while the base TRF in Fig. 6 reflects 1990. In 1990, the temperature at which the TRF reaches its minimum is about 18.5 °C (65.3 °F), very close to the traditional threshold of 65 °F used to define heating and cooling degree days. A 10-degree increase from 5 °C to 15 °C decreases per capita electricity consumption by about 50%, while a 10-degree increase from 19 °C to 29 °C increases per capita electricity consumption by about 80%.

The price TRF (upper right) is plotted such that the vertical axis is the same as the base TRF. We can see that the effect of a standard deviation increase in price is small relative to the effect of temperature itself or relative to the effect of time (bottom left) over a few years or more. We expect a price increase to decrease electricity consumption and this does indeed occur at relatively cold temperatures with statistical significance, but not at relatively hot temperatures, where the effect is statistically insignificant.

The time TRF (bottom left) shows how residential responsiveness to temperature has changed over time. Energy consumption has declined at relatively cold temperatures — but not the coldest temperatures. At relatively hot temperatures, consumption has increased. These changes are both economically and statistically significant. On the surface, these changes contrast with the findings of Chang et al. (2016) of no change over time using the same method but for the Korean residential electricity market. However, those authors remove the level effect of time by detrending consumption data, which could remove some of the interaction of time with temperature that we find for WA.

To get a clearer picture of the effect of time on the TRF, we plot the base TRF against the base TRF plus the time TRF (bottom right). The difference between the two curves is given by the time TRF itself

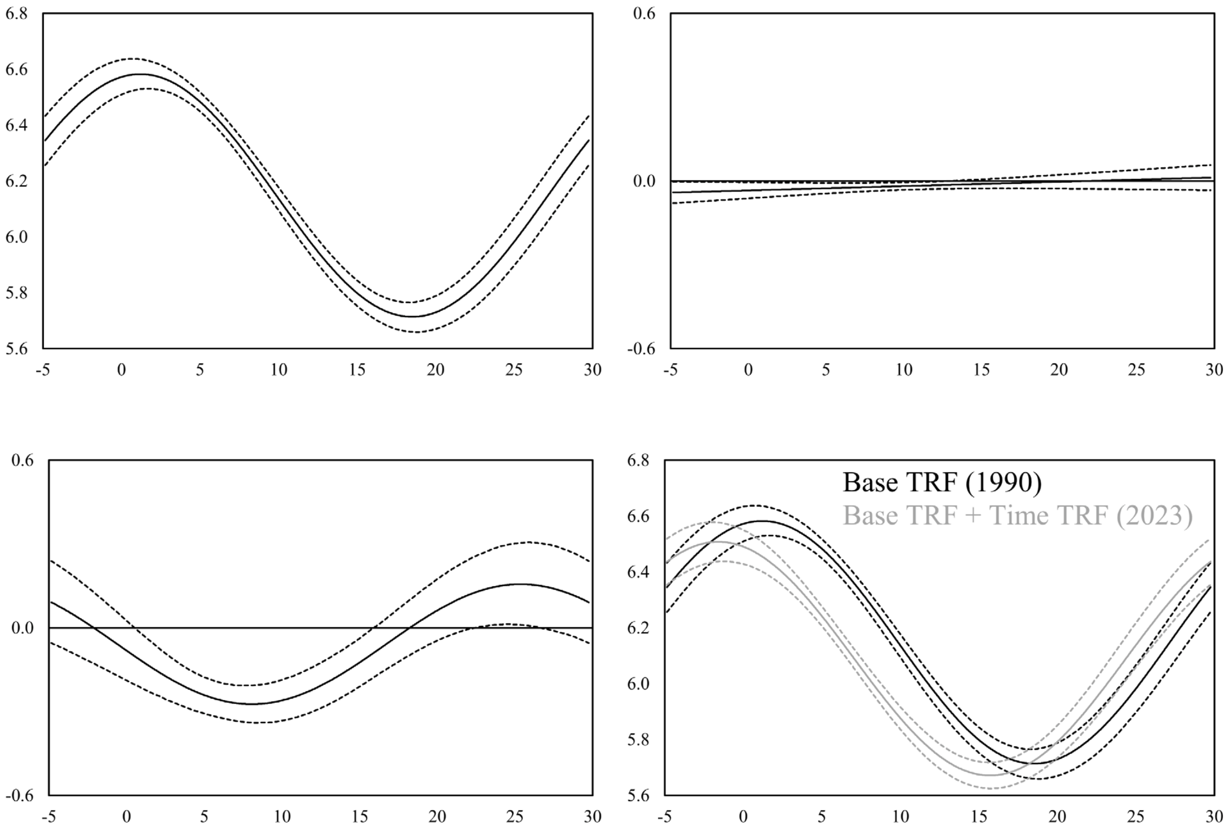

Fig. 6. Residential Consumption: Estimated Base TRF and Cross-TRFs. Unit: log consumption (kWh per capita). Top left panel: base TRF. Top right panel: price TRF such that $p_t = 1$ or one standard deviation more than the sample average price. Bottom left panel: time TRF such that $(t-1)/(T-1) = 1$. Bottom right panel: base TRF plus time TRF. Model estimated from Eq. (15) with parameter estimates given in Table 3. 95% confidence bands constructed using an autoregressive residual-based bootstrap with 9999 replications.

(bottom left), which we can see is both statistically and economically significant over much of the temperature range. Setting aside the possibility of changes in household size, two plausible explanations for the decrease in consumption at colder temperatures over time are gains in efficiency in heating and a decrease in demand due to cheaper alternatives. The change is likely due to a combination of these factors, and the second seems particularly plausible due to the availability and declining price of natural gas, which is the most natural substitute for heating over a duration long enough that appliances are replaced. The increase in consumption at temperatures hotter than 18.2 °C (64.8 °F) is presumably due to more use of air conditioning for cooling, whether due to a greater availability of air conditioning, a decrease in household size (the same cooling for fewer people in a household), or simply a change in habits.

Fig. 7 plots the base TRF and cross-TRFs for the commercial sector. The effects of introducing the cross-TRFs are similar to those in the residential sector. Specifically, the base TRF (for 1990, top left) appears to be similar to the (average) TRF from the model with no cross-TRFs (Fig. 5). Like that for the residential sector, retail price for the commercial sector has a small negative effect over colder temperatures and no statistically significant effect over hotter temperatures (top right).

The time TRF (bottom left) is plotted with the addition of both β^{*}_{100} and β^{**}_{100}, because both $\mathbf{1}_t(t \geq 2002.11)$ and $\mathbf{1}_t(t \geq 2002.11)((t-1)/(T-1))$ equal one at the end of the sample, when $t = T$. Although the shape of the time TRF for the commercial sector is different from that of the residential sector, both are negative at relatively colder temperatures and positive at relatively hotter temperatures. Combining the base and time TRFs (bottom right) shows that energy consumption has increased over time at nearly every temperature down to about 5 °C, below which

Table 4
Decompositions of hourly temperatures using ENSO proxies.

Coeff.	Hourly temperature (°C)		
SST	0.419 (0.009)		
SST^{+}		0.649 (0.019)	
SST^{-}		−0.271 (0.019)	
El Niño			0.898 (0.019)
La Niña			−0.168 (0.019)

SST denotes the anomalies given by the Niño 3.4 series, while $SST+$ and $SST-$ denote the absolute value of the positive and negative anomalies the same series. El Niño and La Niña are binary variables triggered by months in the respective states (see Section 3). Over the sample period of 408 months, there were 96 (23.5%) and 104 (25.5%) months marked by El Niño and La Niña, respectively. All models contain an intercept and eleven monthly dummies. Standardized temperatures in the paper are decomposed using a coefficient of $\hat{\gamma}_E = 0.419/(b-a) = 0.419/(29.750 + 4.855) = 0.012$.

the difference does not appear to be statistically significant. Again, the difference between the two curves is given by the time TRF itself (bottom left), which we can see is both statistically and economically significant above roughly 10 °C.

4.3. Temperature decomposition and counterfactual analysis

Recall that our first step in evaluating economic damage from ENSO is to decompose temperature into components that are correlated and uncorrelated with ENSO, given by the regression in Eq. (12). We proxy ENSO by the series of sea surface temperature anomalies E_t given by

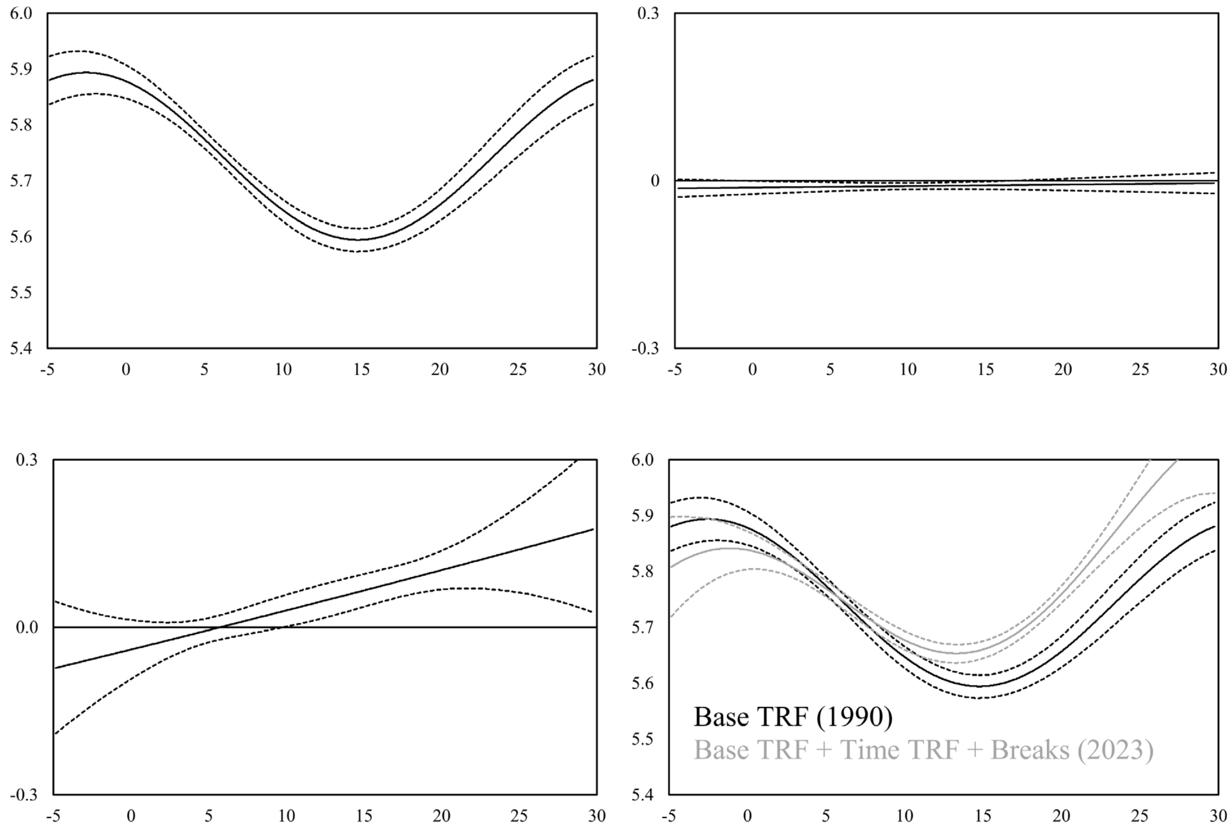

Fig. 7. Commercial Consumption: Estimated Base TRF and Cross-TRFs. Unit: log consumption (kWh per capita). Top left panel: base TRF. Top right panel: price TRF such that $p_t = 1$ or one standard deviation more than the sample average price. Bottom left panel: time TRF such that $(t-1)/(T-1) = 1$ and $\mathbf{1}_t(t \geq 2002.11) = 1$. Bottom right panel: base TRF plus time TRF. Model estimated from Eq. (15) with parameter estimates given in Table 3. 95% confidence bands constructed using an autoregressive residual-based bootstrap with 9999 replications.

Niño 3.4. The regression result shown in the first column of Table 4 gives $\hat{\gamma}_E = 0.419/(b-a) = 0.419/(29.750+4.855) = 0.012$. The table also shows similar regression results from parsing the data into separate series of indicators for positive and negative anomalies and for defined El Niño and La Niña episodes. Not surprisingly, the SST anomaly is positively correlated with WA temperatures. Interestingly, the correlation is strong during hotter periods (including El Niño) and not as strong during colder periods (including La Niña). Compared to months with no El Niño or La Niña, which we will call *no-ENSO* months, hourly temperatures during El Niño episodes are 0.898 °C hotter, while those during La Niña episodes are 0.168 °C colder.

Setting the SST series equal to zero or turning off ENSO indicators switches the temperature conditions during ENSO episodes to temperature conditions during no-ENSO months, so that the counterfactual temperatures are given by $r_0 = r_* - r_E$. Recall from Section 2.2 that this difference occurs on a monthly basis and that r_E is estimated by $E_t'\hat{\gamma}_E$. Fig. 8 shows r_E over each month since 1990. Positive spikes strongly overlap with El Niño episodes (yellow background) and negative spikes strongly overlap with La Niña episodes (blue background). Indeed, the figure is quite similar to the ENSO series itself shown in Fig. 3 in Section 3 but with a smaller vertical scale.

Fig. 9 shows our estimates of economic damages (costs) or negative damages (savings) from ENSO calculated as $\nabla_E w_t' g_{*t}$ as explained in Eq. (7) in Section 2.1. The vertical scale is in logs, so we may interpret increments to approximate percentage changes in consumption per capita. The effect of ENSO in the commercial sector is 2.1% at most, while it is larger, up to 6.4% in the residential sector. Because the signs are nearly always equal across the two sectors, the maximum total effect is indeed given by the sum 8.5%, and this appears from the figure to be a saving from the super El Niño of 1997–1998.

As Fig. 9 suggests, costs associated with ENSO are positively correlated (0.64) with La Niña episodes, while savings are positively correlated (0.61) with El Niño episodes. The literature commonly associates ENSO-induced economic damages with El Niño rather than with La Niña. In an evidently warming environment, the additional warming of El Niño could indeed drive additional damages in hotter climates. Moreover, the moisture transport associated with El Niño may drive destructive natural disasters. However, we have seen that the temperature effects of El Niño are more common in the heating season in WA and in the PNW more generally. Additional environmental heat from El Niño saves rather than costs additional expenditures during the heating season. Similarly, La Niña episodes more often occur during spring months, still the heating season, so that these episodes result in costs rather than in savings.

Looking closely at one particular period, the super El Niño of 2015–2016, provides insights. This episode actually started in October 2014, with an excess temperature due to ENSO of 0.2 °C, which created a savings of 0.7% that month. The next seven months of excess temperature generated savings of 2.3% (Nov), 2.2% (Dec), 1.8% (Jan), 1.9% (Feb), 1.3% (Mar), 2.4% (Apr), and 0.8% (May). However, the excess temperature of June, July, and August of 2015 had the opposite effect, driving cooling measures and costing 2.1%, 3.7% and 3.9%, a shorter but sharper difference from the no-ENSO counterfactual. In hotter parts of the world, the costs associated with the excessive heat of 2015 were much higher. The next nine months, through May 2016, saw high savings, almost as much as 7% during some months. The cumulative effect of this episode was a saving of 2.1% per month or a total of 41.8% of a month spread over 20 months.

Table 5 breaks down average monthly costs or savings of all El Niño and La Niña episodes across both sectors during the sample period.

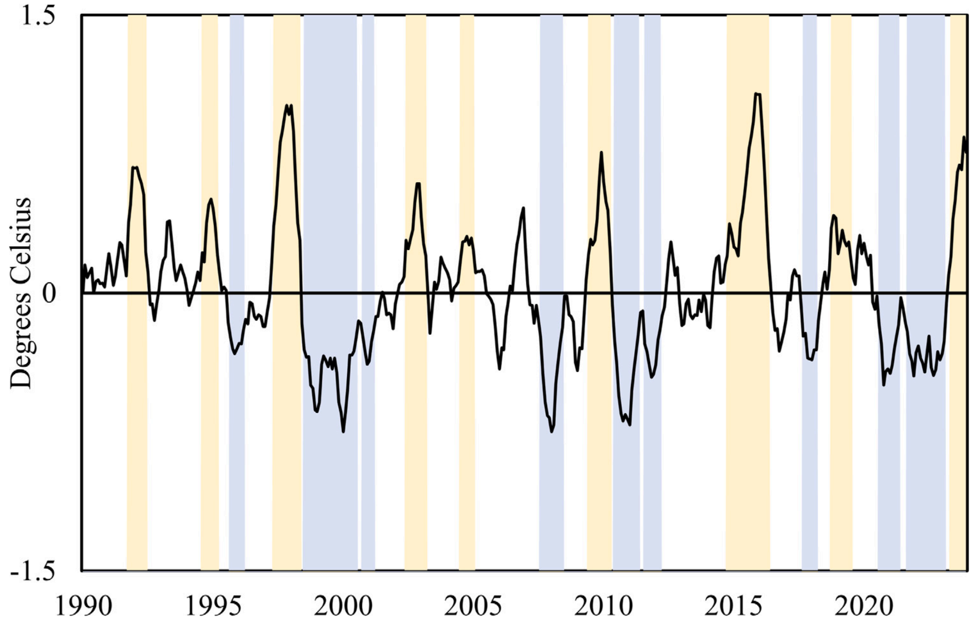

Fig. 8. Excess Temperatures from ENSO. Monthly actual less counterfactual temperatures, $r_E = E_t' \hat{\gamma}_E$. Yellow and blue backgrounds denote El Niño and La Niña episodes, respectively.

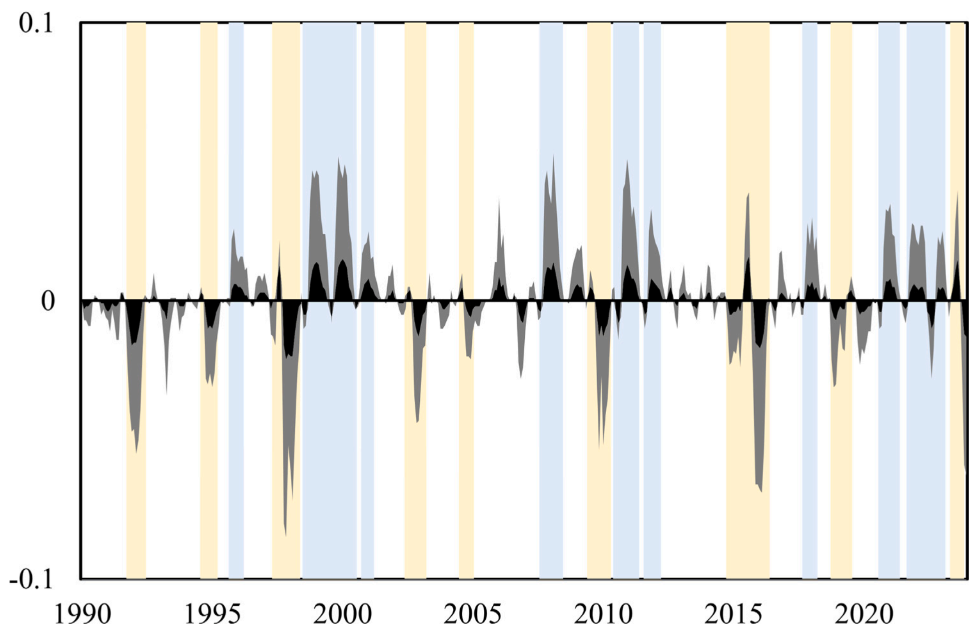

Fig. 9. Electricity Costs and Savings from ENSO. Unit: log consumption (kWh per capita). Economic damages attributable to ENSO as calculated in Eq. (7) using the residential and commercial cross-TRFs with coefficient estimates given in Table 3. Positive values are costs, while negative values are savings. Black denotes costs or savings to the commercial sector, while gray denotes (additional) costs or savings to the residential sector. Yellow and blue backgrounds denote El Niño and La Niña episodes, respectively.

Looking again at the super El Niño of 2015–2016, the total monthly saving of 2.1% is given as the sum across the columns at that row, $0.3\% - 2.1\% + 0.2\% - 0.5\% = -2.1\%$. The super El Niño of 1997–1998 had an even larger monthly saving (3.5%) as did the earlier episode of 1991–1992 (3.6%). The most damaging La Niña episodes were in 2007–2008 (2.7%) and 1998–2000 (2.5%).

Stepping back from specific episodes, if we average all of the costs and savings shown in Fig. 9, we get a net saving of less than 0.001% per month. Our results on savings complement rather than contradict the extensive results on damages from ENSO in the literature, because the effects of ENSO are widely known to be geographically heterogeneous. Reducing electricity consumption in Seattle, Washington due to a mild El Niño boreal winter does not contradict storm damage in Quito, Ecuador or Lima, Peru from a moist El Niño austral summer.

5. Conclusion

We propose a simple regression-based way to assess climate impacts by decomposing noisy temperature fluctuations to identify a component attributable to anthropogenic or natural forcings or natural variability. By doing so, we can generate counterfactual temperatures and hence counterfactual electricity consumption. We view electricity consumption as a “damage” (more appropriately, a cost), and we use a cross-TRF

Table 5
Average monthly electricity costs and savings from ENSO.

El Niño	Residential		Commercial		Total
Duration	Cost	Saving	Cost	Saving	
Oct 1991–Jun 1992	0.000	−0.027	0.000	−0.009	−0.036
Aug 1994–Mar 1995	0.000	−0.014	0.001	−0.006	−0.019
May 1997–May 1998	0.001	−0.029	0.002	−0.009	−0.035
Jun 2002–Mar 2003	0.000	−0.015	0.001	−0.005	−0.019
Jul 2004–Jan 2005	0.001	−0.009	0.001	−0.002	−0.009
Jun 2009–Apr 2010	0.001	−0.018	0.001	−0.006	−0.022
Oct 2014–May 2016	0.003	−0.021	0.002	−0.005	−0.021
Oct 2018–Jul 2019	0.001	−0.012	0.001	−0.003	−0.013
May 2023–Dec 2023	0.006	−0.015	0.005	−0.003	−0.007
El Niño Average	0.001	−0.018	0.002	−0.005	−0.020
La Niña	Residential		Commercial		Total
Duration	Cost	Saving	Cost	Saving	
Sep 1995–Mar 1996	0.012	−0.000	0.004	−0.000	0.016
Jul 1998–Jul 2000	0.020	−0.000	0.006	−0.001	0.025
Oct 2000–Mar 2001	0.014	−0.000	0.006	−0.000	0.020
Aug 2007–Jun 2008	0.021	−0.000	0.007	−0.001	0.027
Jun 2010–May 2011	0.019	−0.001	0.005	−0.002	0.021
Aug 2011–Mar 2012	0.013	−0.001	0.004	−0.001	0.015
Sep 2017–Mar 2018	0.015	−0.000	0.004	−0.000	0.019
Aug 2020–May 2021	0.015	−0.001	0.004	−0.001	0.017
Sep 2021–Feb 2023	0.013	−0.002	0.003	−0.002	0.012
La Niña Average	0.016	−0.001	0.005	−0.001	0.019
ENSO Average	0.009	−0.009	0.003	−0.003	−0.000

Unit: log consumption (kWh per capita). Economic damages attributable to ENSO as calculated in Eq. (7) using the residential and commercial cross-TRFs with coefficient estimates given in Table 3. Positive values are costs, while negative values are savings.

modeling approach that is nonlinear yet simple to implement using linear regression and that is novel to the climate impact literature to the best of our knowledge.

We specifically examine the state of Washington, the northwestern-most of the contiguous 48 states of the US, a region with temperatures that are know to fluctuates with the El Niño Southern Oscillation. The cross-TRFs reveal time and price effects on the demand response to temperature. Both residential and commercial sectors appear to have become more frugal at low and intermediate temperature and less frugal at high temperatures since 1990. We speculate that the frugality of the heating side results from a diversification of energy sources used for heating, while the lack of frugality on the cooling side results from an increasing number of air conditioners installed in a traditionally temperate zone. Price effects in both sectors appear to be small and significant only over a limited range of temperatures.

Though our methodology is quite general, we are motivated by the effect of the El Niño Southern Oscillation on electricity consumption, which appears to be an understudied effect of ENSO. Because the effects of ENSO are primarily felt in the heating season in this region *and* the colder La Niña episodes tend to happen during colder seasons than hotter El Niño episodes, the net effect of ENSO is a negative "damage" or an electricity *saving*. This empirical result is likely specific to this region, and we would expect a net cost of ENSO in hotter regions. Moreover, we must emphasize that the saving is with respect to consumption of energy but we do not consider ways in which ENSO may have damaging effects already well-documented in the literature, such as flooding, drought, or mortality or morbidity from excess heat or cold exposure.

Our approach to identifying an ENSO component of temperature is simple and incorporates any linear contemporaneous correlation of ENSO and temperature. However, its simplicity may be a limitation in that more complicated nonlinear dynamics or interactions are not captured explicitly. Careful consideration of this limitation would be warranted if the methodology were used for forward-looking analyses such as scenario-based forecasts. For example, if the frequency and intensity of ENSO is a function of anthropogenic forcing, as suggested by Cai et al. (2015, 2018), that relationship would be an important key to projecting ENSO-induced damages into the future.

If we were to scale up our analysis to a global level, our implicit assumption on the exogeneity of ENSO would need to be examined more carefully. As a component of natural variability of the climate system, ENSO and other cycles are not forcings, but may themselves be forced, as in the case of anthropogenic forcing affecting ENSO mentioned above. But forcings themselves need not be exogenous to the climate system: economic impacts of climate change may drive emissions through physical damages or policy, which in turn force the climate system. While such feedback loops are important, our focus on a single region or a "small" agent in the economic sense allows us to treat ENSO as exogenous.

CRediT authorship contribution statement

J. Isaac Miller: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Fangyu Zhong:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eneco.2026.109301.

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